



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 39 summarizes the current fairness state for upload 39. Accuracy changed to +0.787 and fairness score changed to +0.955. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.888.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.787 ▲ +0.787	0.955 ▲ +0.955	0.888 ThresholdOptimizer_mitigated_threshold	0.823 Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	compas_dt_baseline.pkl	comparison

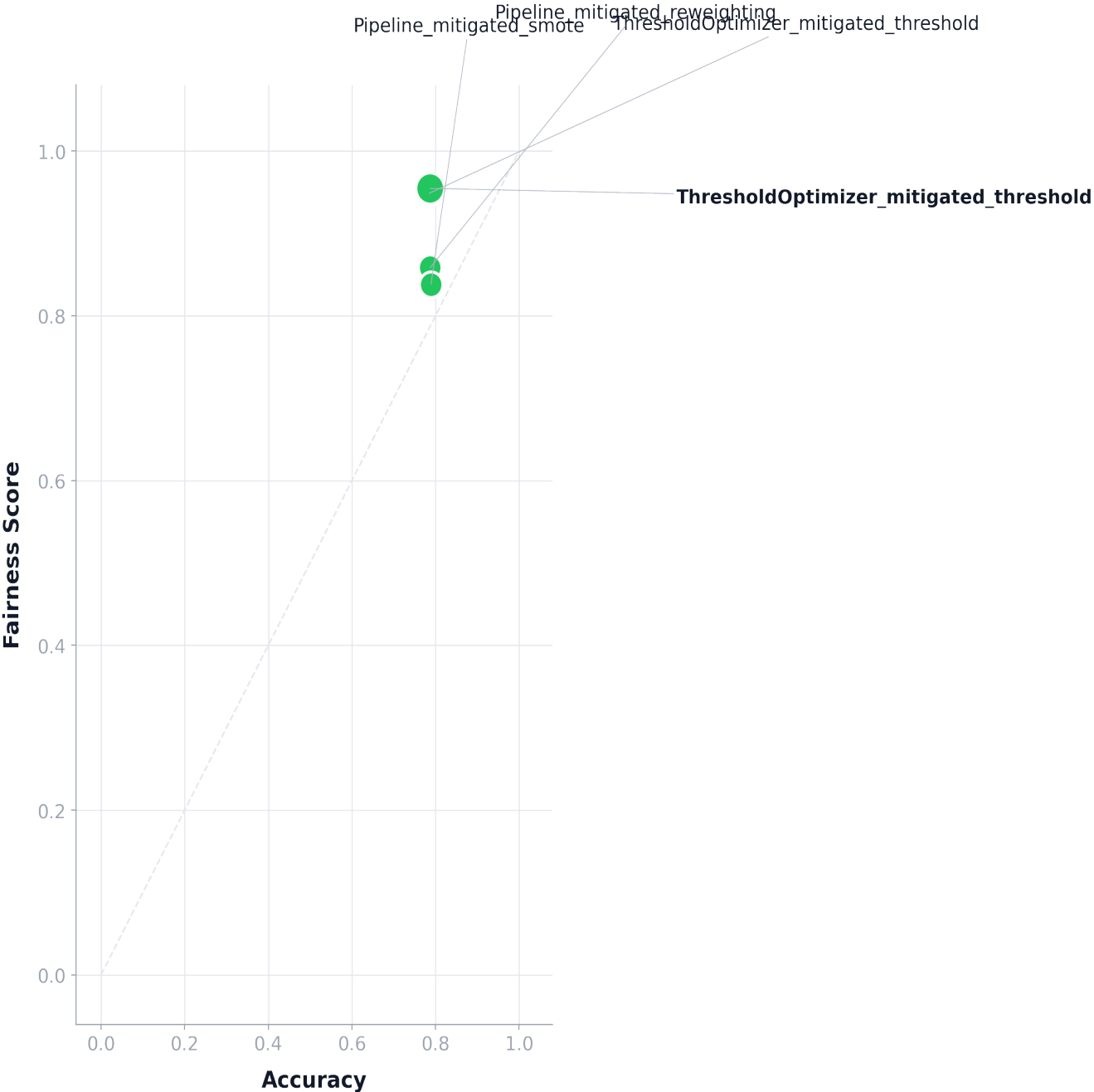
Fairness Metrics

METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.080	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.011	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.823	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.787	0.955	0.080	0.011	0.823	0.888
★ ThresholdOptimizer_mitigated_threshold	0.784	0.948	0.083	0.021	0.816	0.883
Pipeline_mitigated_reweighting	0.787	0.858	0.166	0.117	0.633	0.830
Pipeline_mitigated_smote	0.789	0.838	0.185	0.139	0.609	0.818

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.